

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech. (CL) Examination****April - May, 2018****CL305 Environmental Engineering II****Date: 03.04.2018, Tuesday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70*****Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below: [07]**

- (a) Draw a layout of conventional sewage treatment plant. Write the function of each unit. [05]
- (b) How temperature affects quality of sewage? [02]

Q - 2(a) Explain environmental impact of waste generation. [04]

- (b) Find diameter of circular primary settling tank for flow $0.35 \text{ m}^3/\text{s}$. Take surface loading rate = $40 \text{ m}^3/\text{m}^2/\text{day}$. [05]

OR

- (b) Design a grit chamber for population 50000 and water consumption 135 lit/capita/day. [05]
- (c) What do you mean by suspended and attached growth processes? Give example of each. [05]

Q - 3 Attempt any Two: [14]

- (a) Enlist various physical, chemical and biological characteristics of domestic wastewater. Differentiate domestic and industrial wastewater.
- (b) Explain with neat sketch biological process occurring in a trickling filter.
- (c) Draw a neat graph of BOD v/s incubation period. Derive a first-order equation for BOD remaining at any time t .

SECTION – II

Q - 4 Answer the questions: [07]

- (1) Distinguish between rubbish and garbage. [02]
- (2) Distinguish between stationary and mobile sources of air pollutants. [02]
- (3) What is meant of smog? Discuss its causes and effects. [03]

Q - 5(a) Describe how the refuse is collected, removed and transported. [05]

(b) What is particulate matter? How particulate matters harms human health. [05]

OR

(b) Write a detailed note on incineration of refuse. What are its advantages and disadvantages? [05]

(c) Write short note on EIA. [04]

Q - 6(a) Write detailed not on composting. [07]

(b) Write a note on control of gaseous pollutants. [07]

OR

(b) Enumerate various types of equipments used for control of particulate matter and explain any one in details with neat sketch. [07]
